

Problems Proposed for 2025-26 Pal4Pal

Tianlin, Franklin, Vincent

August 2025

1 Tianlin

1.0.1 P1 Algebra [4 points]

What is the remainder when $2^{202} + 202$ is divided by $2^{101} + 2^{51} + 1$?

1.0.2 P2 Geometry [5 points]

Isosceles trapezoid $ABCD$ has parallel sides $\overline{AD}$ and $\overline{BC}$, with $BC < AD$ and $AB = CD$. There is a point P in the plane such that $PA = 1, PB = 2, PC = 3$, and $PD = 4$. What is $\frac{BC}{AD}$?

2 Franklin

2.0.1 P1 Combo [4 points]

Fred the Frog starts at $(0,0)$ of the coordinate plane. He wishes to get to $(4,4)$, and each turn he may move one unit right, one unit left, or one unit up. Given that he may never visit the same grid space twice and for any point (x,y) that he visits, $0 \leq x, y \leq 4$. how many ways are there for him to reach the said location?

2.0.2 P2 Combo [4 points]

Alice, Bob, and Carol are in a math relay tournament. Due to poor planning and strategy, each person does not work on the problem until the person before them has finished (Alice starts, Bob starts once Alice is complete, and Carol starts once Bob is complete). Each of them takes a random time between 0 and 5 minutes to solve the problem. Given that there is a 5-minute total time limit, what is the probability that Carol can finish the problem?

3 Vincent

3.0.1 P1 Combo [4 points]

Alice has 5 friends and they are sitting in a circle and passing a ball around. The ball starts in Alice's hands. What is the probability that Alice ends up with the ball after they pass it 7 times? People cannot pass to themselves or to people outside the circle. (when they are passing the ball around, the person with the ball gives it to someone else in the group with equal probability. This counts as a turn, so when passing it 7 times, the ball changes holder 7 times, eg. Alice – Bob – Alice – Bob – Alice – Bob – Alice – Bob is 7 turns.)

3.0.2 P2 Combo [4 points]

Bob is driving home from work. Suppose Bob's work is located at point $(0, 0, 0)$ and his home is at $(6, 6, 6)$. However, Bob is hungry and wants to go to a burger store. The burger store is located at $(2, 1, 3)$. Bob has little gas, so he wants to use the shortest route through the burger store home. However, along the way, there is a gang at $(4, 3, 5)$, and Bob should avoid this gang. If Bob can only move through consecutive lattice points, how many ways are there for him to get back home?